MDT904: Research and Design Thinking for Digital Transformation

Assignment 1 – Literature Review

Due: October 25th, 2025

Assessment Task:	**Literature review**
Type	Individual
Learning Outcome Measured	1, 2
Total Marks:	100
Weighting:	30%
Due Date:	Session 4 – October 25th, 2026

Description:

In this assignment, students will be given a business use case, and they need to conduct a literature review to identify the existing solutions proposed so far, their strengths and limitations, and the gaps that still exist in the literature. The literature review should highlight the role that research and innovation is used the address the problem at hand. A minimum of 6 scholarly articles should be analyzed.

Marking Criteria:

- In-depth analysis of relevant papers [20%]
- Identification of research gaps [30%]
- Quality of references [20%]
- Quality of the presentation [20%]

Business Case Title: Enhancing Customer Experience in E-commerce through AI-Powered Personalization

Background: In today's highly competitive e-commerce landscape, providing a personalized and seamless customer experience is paramount. E-commerce businesses are increasingly turning to artificial intelligence (AI) and machine learning (ML) to tailor their services to individual customer preferences. This business case explores the need for AI-powered personalization in e-commerce and the role of research and innovation in addressing this need.

Problem Statement: Many e-commerce platforms struggle to provide a personalized shopping experience for their customers, resulting in missed sales opportunities and reduced customer satisfaction. The challenge lies in harnessing AI and ML technologies effectively to offer personalized product recommendations, content, and user interfaces.

Assignment Objective: The objective of this assignment is for students to conduct a comprehensive literature review on the role of research and innovation in addressing the challenge of AI-powered personalization in e-commerce. They are expected to identify existing solutions, analyze their strengths and limitations, and critically assess the gaps in the current research landscape.

Assignment Details:

1. **Literature Review:** Students will need to identify a minimum of 6 scholarly articles that focus on AI-powered personalization in e-commerce. These articles should cover a range of topics, including recommendation algorithms, customer profiling, user interface personalization, and the impact of AI on customer experience.
2. **In-depth Analysis:** For each selected article, students should provide an in-depth analysis, summarizing the key findings, methodologies used, and the relevance of the research to the business problem. They should critically evaluate the strengths and limitations of the proposed solutions.
3. **Identification of Research Gaps:** Students should pay special attention to identifying research gaps in the existing literature. What aspects of AI-powered personalization in e-commerce remain unexplored or underdeveloped? Are there emerging trends or technologies that could enhance the current state of research?
4. **Quality of References:** Students are expected to use high-quality, peer-reviewed scholarly articles and reputable academic sources in their literature review. They should demonstrate their ability to select relevant and credible references.
5. **Quality of Presentation:** The literature review should be presented clearly and coherently, following academic writing conventions. Proper citation and referencing should be observed, and the structure of the review should flow logically, making it easy for readers to follow the argument.

Expected Deliverables:

- A literature review document that summarizes and critically analyzes a minimum of 6 scholarly articles related to AI-powered personalization in e-commerce.
- A report that highlights the key findings, research gaps, and the role of research and innovation in addressing the identified problem.
- **Submission should be done using the provided Moodle Link. Submission deadline is October 25th, 2026 – Midnight (UAE time).**
- Late submission: 20% per day penalty